AI powered speech-to-speech mock interview platform

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The study examines the project from four angles — technical, economic, operational, and legal/ethical feasibility — and concludes with an overall assessment of viability

1. Technical Feasibility

Technical feasibility assesses whether the required technology exists, is accessible, and can realistically be implemented by the team given their skills and available hardware.

- All core components (speech-to-text, text-to-speech, resume parsing, LLM-based question generation) rely on mature, freely available, open-source or free-tier tools — faster-whisper, Piper, PyMuPDF, and Ollama or Groq. None require building novel algorithms from scratch.

- The target hardware is sufficient to run a quantised local LLM (3B–8B parameters), local speech-to-text, and local text-to-speech, though with some latency that the team must plan around.

- Each required skill (Python backend development, React/Next.js frontend development, prompt engineering, basic audio handling) falls within standard computer science coursework and is achievable without specialised prior expertise.

- The riskiest technical component is pipeline latency — chaining speech-to-text, LLM inference, and text-to-speech sequentially can introduce noticeable delay. This is a known, manageable risk rather than a blocking one, addressable through model size selection and optional fallback to faster hosted APIs.

2. Economic Feasibility

Economic feasibility assesses whether the project can be completed within a reasonable budget, which for a student project is effectively at or near zero cost.

- The core MVP stack (Ollama, faster-whisper, Piper, FastAPI, Next.js, SQLite) is entirely free and open-source, requiring no software licensing costs.

- Optional paid fallbacks (Groq free tier, temporary OpenAI/Anthropic credits for demo-day reliability) are inexpensive, typically under $20 total if used at all, and are not required for a working prototype.

- No specialised hardware purchase is required; the project runs on hardware the team already owns.

- Hosting for a public demo, if desired, is available through free tiers (Vercel, Render) at no cost, with the only likely paid step being a custom domain name (~$10–12/year), which is entirely optional.

3. Operational Feasibility

Operational feasibility assesses whether the finished system will actually be usable and useful.

- The target users (engineering students) are comfortable with web-based tools and browser microphone permissions, requiring no specialised training to use the system.
- Because the MVP requires no user accounts and works as a single self-contained session, onboarding friction is minimal — a student can use it without any setup beyond visiting the page and uploading a resume.

4. Legal & Ethical Feasibility

This project handles personal data (resumes) and biometric-adjacent data (voice recordings), which introduces some legal and ethical considerations even at student-project scale.

- Resumes and audio recordings should be stored only as long as necessary for the session and should be deletable by the user, consistent with basic data privacy good practice.
- If any cloud-based API (Groq, OpenAI, etc.) is used, resume and voice data will be transmitted to a third-party service; this should be disclosed to users, and a fully local (Ollama/faster-whisper/Piper) mode should remain available for privacy-sensitive users.
- The system does not currently include facial or video analysis, avoiding the more significant biometric privacy concerns associated with commercial video-interview platforms.
- As an academic project evaluating student-submitted resumes and speech, no special institutional ethics approval is anticipated, but informed consent should be obtained from any test users before their resume or voice data is used for testing or demonstration purposes.